

EE3621: Digital Signal Processing

Lecture 2: LTI Systems, Convolution, Stability & Difference Equations (III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Review of System Properties & Introduction to LTI Systems.
 - **05:00 – 15:00 (10 mins):** Derivation of the Convolution Sum & Graphical Steps.
 - **15:00 – 22:00 (7 mins):** Properties of Discrete-Time Convolution (with proofs).
 - **22:00 – 30:00 (8 mins):** LTI Causality, BIBO Stability (with proof), and Step Response.
 - **30:00 – 38:00 (8 mins):** Constant-Coefficient Difference Equations: Homogeneous & Particular Solutions.
 - **38:00 – 40:00 (2 mins):** Checkpoints & Quick Q&A.
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2 Derivation of the Convolution Sum

An LTI system is completely characterized by its impulse response $h[n] = \mathcal{T}\{\delta[n]\}$. By the sifting property, any input $x[n]$ is expressed as:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \quad (1)$$

Applying linearity and time-invariance of $\mathcal{T}$:

$$y[n] = \mathcal{T}\{x[n]\} = \sum_{k=-\infty}^{\infty} x[k]\mathcal{T}\{\delta[n-k]\} = \sum_{k=-\infty}^{\infty} x[k]h[n-k] \quad (2)$$

This is the convolution sum, denoted as $y[n] = x[n] * h[n]$ (see Figure 1).



Figure 1: Step-by-step graphical progression of convolution: folding, shifting, multiplication, and summation.

3 Properties of Discrete-Time Convolution

3.1 Commutativity

$$x[n] * h[n] = h[n] * x[n] \quad (3)$$

Proof: Let $m = n - k \Rightarrow k = n - m$.

$$x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{m=-\infty}^{\infty} x[n-m]h[m] = h[n] * x[n] \quad (4)$$

3.2 Associativity and Distributivity

$$(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n]) \quad (5)$$

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n] \quad (6)$$

3.3 Shifting Property

If $y[n] = x[n] * h[n]$, then:

$$x[n - n_1] * h[n - n_2] = y[n - n_1 - n_2] \quad (7)$$

4 Causality, BIBO Stability & Step Response

4.1 Causality

An LTI system is causal if and only if:

$$h[n] = 0 \quad \forall n < 0 \quad (8)$$

4.2 BIBO Stability

An LTI system is BIBO stable if and only if its impulse response is absolutely summable:

$$S = \sum_{n=-\infty}^{\infty} |h[n]| < \infty \quad (9)$$

Proof: If $|x[n]| \leq M_x$, then:

$$|y[n]| \leq \sum_{k=-\infty}^{\infty} |x[k]| |h[n - k]| \leq M_x \sum_{m=-\infty}^{\infty} |h[m]| = M_x S < \infty \quad (10)$$

Stability profiles are shown in Figure 2.

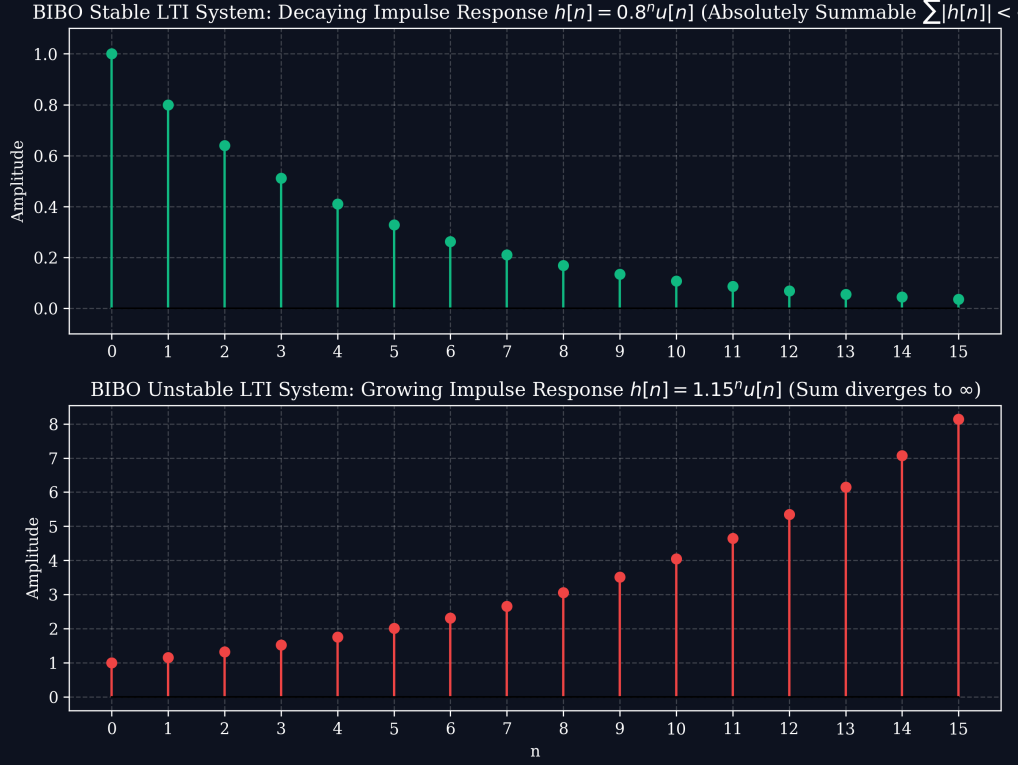


Figure 2: Impulse response profiles for stable and unstable discrete systems.

4.3 Step Response

The unit step response is defined as:

$$s[n] = h[n] * u[n] = \sum_{k=-\infty}^n h[k] \quad (11)$$

The impulse response is recovered by:

$$h[n] = s[n] - s[n-1] \quad (12)$$

5 Analytical Convolution Example (Infinite Sequences)

Let $x[n] = a^n u[n]$ ($0 < a < 1$) and $h[n] = u[n]$.

$$y[n] = \sum_{k=-\infty}^{\infty} a^k u[k] u[n-k] \quad (13)$$

For $u[k]u[n-k] \neq 0$, we require $0 \leq k \leq n$.

- For $n < 0$: $y[n] = 0$.
- For $n \geq 0$:

$$y[n] = \sum_{k=0}^n a^k = \frac{1 - a^{n+1}}{1 - a} \quad (14)$$

Thus: $y[n] = \left(\frac{1 - a^{n+1}}{1 - a} \right) u[n]$.

6 Constant-Coefficient Difference Equations

Discrete-time LTI recursive systems are described by difference equations:

$$y[n] + \sum_{k=1}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k] \quad (15)$$

The total solution is $y[n] = y_h[n] + y_p[n]$.

6.1 Homogeneous Solution $y_h[n]$

Set $x[n] = 0$, yielding:

$$y_h[n] = C\lambda^n \implies \lambda^N + a_1\lambda^{N-1} + \dots + a_N = 0 \quad (16)$$

For distinct roots λ_k , the homogeneous solution is:

$$y_h[n] = \sum_{k=1}^N C_k \lambda_k^n \quad (17)$$

6.2 Particular Solution $y_p[n]$

Represented by trial functions corresponding to the inputs, e.g., $y_p[n] = K_p$ for constant inputs, or $y_p[n] = K_p\beta^n$ for exponential inputs $x[n] = A\beta^n$.